MIL-B-36408B
29 August 1978
SUPERSEDING
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MILITARY SPECIFICATION

BOX, PATHOLOGICAL SPECIMEN SHIPPING, INSULATED, Plastic

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of plastic insulated box, suitable for storage and shipment of frozen or refrigerated medical specimens and biologicals.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated, and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-W-530 - Webbing, Textile, Cotton, General Purpose, Natural or in Colors.

STANDARDS

FEDERAL

FED-STD-751 - Stitch, Seam, and Stitching.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8115

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

20759 - Box, Pathological Specimen Shipping, Insulated, Plastic, 2 cu ft.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Insulation. The insulation for the box and lid shall be fabricated from molded expanded polystyrene - foam plastic. The polystyrene - foam shall have a density of 2-1/2 ± 1/4 pounds per cubic foot, and a maximum thermal conductivity (K factor) of 0.23 B.T.U./ sq. ft./ hr./ ^{0}F./ in. -- at a temperature of 75^{0} Fahrenheit. The insulation for the box shall be acid-and alkaline-resistant.

* 3.1.2 Fiberboard. The outer box of the fiberboard shall be a water-resistant class, solid, fiberboard laminate conforming to type SF, grade W5s, of PPP-F-320. The outside of the fiberboard box shall be an off-white marble color and shall have an overall minimum thickness of 0.075 inches. The laminates comprising the overall solid board shall be bonded with a water-resistant adhesive resin, and the final fiberboard product shall have a basis weight of 310 ± 5 pounds. The weatherproof-treated fiberboard shall be moisture-and water-resistant, and shall not delaminate.

3.1.3 Carrying strap harness.

* 3.1.3.1 Webbing. The webbing for the carrying strap harness shall be black in color, conforming to MIL-W-530 for type IIa or type III, class 4, 1-1/2 inches wide. The maxim copper content on the webbing from copper-8-quinolinolate shall not exceed 1.28 percent.

* 3.1.3.2 Buckle. The buckles for the straps shall be two-bar tongueless types, fabricated from a suitable metal, and shall have a black finish that is not tacky.

* 3.1.3.3 End clips. The end clips for the strap shall be fabricated from a suitable metal and shall have a black finish that is not tacky.

3.1.3.4 Thread. The thread for securing the harness strap webbing shall conform to V-T-276 for type IB3.

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3.2 Construction. The pathological specimen shipping box shall consist of an insulated container encased with a laminate fiberboard, with a two-handled carrying strap. The components shall be fabricated and assembled in accordance with Drawing 20759. The assembled specimen shipping box shall have a weight (empty) of 13 ± 1 pound.

* 3.2.1 Insulated container. The insulated container shall consist of a rectangular body and a detachable plugged lid. Both components shall be of one-piece molded construction. All inside and outside surfaces of the body and lid shall have a molded skin surface finish. When the lid is secured to the body, the plug section shall form a snug fit with the inside walls of the body. The container shall have a minimum load capacity of 75 pounds. When tested as specified, the container in the upright position shall be leakproof, and shall show no evidence of leakage (see 4.4.1).

3.2.2 Fiberboard box. The fiberboard shall encase the insulated container as indicated in Drawing 20759. The inside bottom of the laminated fiberboard case shall be chemically bonded to the outside bottom of the foam-plastic box with a bonding agent (adhesive) compatible with the foam-plastic and fiberboard without deleterious effects on either material. The lid shall extend over the sides of the box so as to make a close fit with the top of the fiberboard case without undue binding. Joints shall be firmly secured with staples suitable for the intended purpose, as shown on the drawing.

3.2.3 Carrying strap harness. The carrying strap harness shall consist of two straps with carrying handles. As indicated in Drawing 20759, one strap shall fit around the height of the specimen shipping box, and the other strap shall fit around the girth of the specimen shipping box. Each strap shall have a metal buckle (see 3.1.3.2) that is secured in a loop formed by the webbing of the strap. The loop shall be fastened by box tack sewing as indicated in Drawing 20759. The raw end of the loop, to prevent raveling, shall either be turned under and double stitched, or impregnated with a hardening material. The opposite end of the strap shall terminate with an end clip specified in 3.1.3.3. The webbing shall be inserted the full depth of the end clip and shall be flat at the point of entry. The end clip shall be firmly attached without cutting the webbing. The straps shall be attached to each other as indicated in Drawing 20759, to provide a one-piece sling harness. Handle configurations, made of the same webbing material, shall be attached at points above and below the junction points of the straps. Attachment points of the sling strap harness (strap cross points, handle configurations, and buckle-securing points) shall be accomplished by box tack sewing. The thread shall conform to 3.1.3.4. All stitching shall conform to stitch type 301 of Federal Standard No. 751, with six to eight stitches per inch. Thread tension shall be maintained so that there is no loose stitching, and the lock shall be embedded in the webbing material. The straps shall be of suitable length so that the girth of the container, with the buckles located approximately halfway across the cover of the closed container and approximately halfway across the girth of the container. The clip ends of the buckled straps shall extend through the buckles with a minimum gripping length of six inches.

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* 3.3 Identification marking. The name or registered trademark of the manufacturer shall be permanently and legibly marked on the exterior surface of the bottom of each fiberboard box.

* 3.4 Workmanship. Workmanship shall be first class throughout. The boxes and harness straps shall be free from defects which detract from their appearance or impair their serviceability. The end clips on the harness straps shall pass freely through the buckles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the fiberboard, webbing, and thread, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of inspection shall be the assembled specimen shipping box with carrying strap harness.

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TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. For sampling.

Component (unit of product)	Characteristic	Require-ment	Test procedure	Inspection level	AQL
Plastic box only	Leakage	3.2.1	4.4.1	S-2	*

*Acceptance number shall be zero.

* 4.3 Examination. The box shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III, classification of defects.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Plastic box and lid not free of chips, cuts, and indentations.
102	Plastic lid does not fit snugly in foam box.
103	Fiberboard box and lid not free of deformation.
104	Fiberboard box and lid not free of tears, cuts, holes, or indentations.
105	Sewing on strap harness not free of loose stitching.
106	Lid of fiberboard box does not fit closely to top of box without undue binding.
107	Plastic body and lid not free of crushed edges or gouges.
108	Fiberboard box or lid not free of crushed, ragged, or uneven edges.
109	Stapled sides (corners) of fiberboard body or lid not properly secured (stapling not well-clinched or does not pass through all pieces to be fastened).
110	Webbing straps not free of cuts or holes.
111	Assembled box on strap harness not free of missing component.
112	Clip not adequately or securely attached to webbing of strap.
113	End clip on harness strap not capable of passing freely through buckle.

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TABLE III. Classification of defects. (Cont'd)

Categories	Defects
Minor	
201	Identification marking not complete, not permanent, illegible, or missing.
202	Fiberboard box or lid not free of chipping at edges.
203	Fiberboard box lid not free of scuff or deep scratches.
204	Plastic box not free of foreign matter or loose or ingrained dirt.
205	Fastening staples not free of visible cracking or rust.
206	Webbing not flat at point of entry to clip.
207	Buckle or end clip of harness not free of tackiness.

*Examination shall not be restricted to the classified possible defects listed above.

* 4.3.2 Dimensional examination. The unit box shall be examined for defects in dimensions. Any dimension not within the limits specified herein shall be considered as a defect.

* 4.3.3 Packaging inspection. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leakage. The plastic box shall be set on newsprint or similar type paper and filled with tap water to within 3 inches from the top of box. The water-filled box shall be allowed to stand for 24 hours before examining for evidence of leakage on the paper and for moisture on the sides of the box.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each box shall be clean, dry fully assembled and wrapped in kraft paper or inserted in a plastic bag having a minimum thickness of 2 mil. The wrap or bag shall be secured with tape.

5.1.2 Level C. Each box shall be clean, dry and packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Each box packaged as specified in 5.1.1.1 shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Each box packaged as specified in 5.1.1.1 shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged boxes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform the the carrier's rules and regulations.

5.3 Marking. Marking shall be as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Box, Pathological Specimen Shipping, Insulated, plastic, is intended for use in shipping and storage of medical specimens and biologicals.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number (NSN).

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.4 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
8115-00-226-1199	BOX, PATHOLOGICAL SPECIMEN, SHIPPING, INSULATED, Plastic, 2 cu. ft.

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-B-36408B BOX, PATHOLOGICAL SPECIMEN SHIPPING, INSULATED, Plastic

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